

Atharva Tupe

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PROFESSIONAL SUMMARY

Software Engineer and MS CS candidate experienced in scalable backend systems, real-time applications, and AI-powered products. Skilled in Python, C++, Java, FastAPI, cloud infrastructure, LLMs, and production APIs

EDUCATION

Binghamton University, State University of New York, NY, USA

Expected May 2027

Master of Science in Computer Science | GPA: 3.71/4.0

Coursework: AI/ML, Data Structures & Algorithms, OS, Cloud Computing, Database Systems, Software Engineering

MIT ADT University, Pune, India

May 2025

Bachelor of Technology in Computer Science and Engineering

WORK EXPERIENCE

EasyBee AI, Inc.

Sept 2026 – Present

AI Engineer

Boston, MA

- Re-architected EasyBee AI's production voice agent platform from a cascaded STT → LLM → TTS pipeline to a native speech-to-speech architecture using LiveKit, Deepgram, and LangGraph, enabling lower-latency real-time conversations
- Built an in-house voice activity detection system using LiveKit Flux with a backup VAD architecture, replacing WebRTC and Silero VAD and reducing silence/endpoint detection latency from 0.5s to 0.05s
- Engineered multi-turn conversation awareness and interruption (barge-in) handling within LangGraph-orchestrated workflows, preserving dialogue context across interruptions during live customer calls
- Developed and deployed customer-facing voice agent workflows using Python, FastAPI, Twilio, LiveKit, and AWS for enterprise self-storage clients across the US and UK

Binghamton University

Dec 2025 – Present

AI/ML Graduate Research Assistant

Binghamton, NY

- Fine-tuned BERT and Bidirectional LSTM models in PyTorch and TensorFlow to extract text and link spans from Wikipedia articles, improving model accuracy by 15%
- Applied graph neural networks to predict links between Wikipedia articles, using cross-validation to compare models and select the most reliable approach
- Trained and fine-tuned Hugging Face transformer models and deployed them on Google Cloud Vertex AI, moving research models from experiments into production

Denner Ventures Pvt. Ltd.

Oct 2024 – Mar 2025

Software Development Engineer

Pune, India

- Developed a student housing platform using React and Express.js, building core features for landlord-student communication and roommate matching across the application
- Engineered scalable frontend features and deployed them to production, supporting 1,000+ concurrent users with responsive user experiences across desktop, tablet, and mobile devices

PROJECTS

ResumeIQ | Python, FastAPI, LangChain, GPT-4, RAG, AWS | GitHub

- Built an end-to-end RAG pipeline for JD parsing and semantic retrieval using **LangChain** and **GPT-4**
- Designed orchestration across ingestion and retrieval stages using **Apache Airflow**, **Kafka**, and **Docker** on **AWS**
- Implemented modular microservices with **PostgreSQL** and **Redis** caching behind RESTful APIs

DailyDock | React, FastAPI, PostgreSQL, Vercel | GitHub

- Built a campus dashboard integrating live shuttle, dining, and calendar data for **100+ Binghamton students**
- Reverse-engineered a geolocation-aware transit ETA API (OCCT SPOT API) to power real-time shuttle tracking
- Implemented **Google OAuth** + **JWT**-based authentication with WCAG/ARIA accessible UI design

TECHNICAL SKILLS

- **Languages & Backend:** Python, Java, C++, TypeScript, JavaScript, SQL, FastAPI, Flask, Node.js/Express.js, REST APIs
- **AI/ML & Voice:** PyTorch, TensorFlow, LLMs, RAG, LangChain, LangGraph, Hugging Face, BERT, LiveKit, Deepgram, Twilio
- **Cloud, DevOps & Databases:** AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD, Git/GitHub, PostgreSQL, MongoDB, Redis